POCKETBOT APEX — COMPLETE BEGINNER MANUAL

Zero-Assumption Guide to the Hybrid Dual-Broker (MetaTrader 5 + cTrader) & 3-Tier AI Trading Engine
Written for Complete Beginners — Every Click, Keystroke, Example, Safety Brake, Command & Fix Explained.

1. Welcome: What Is PocketBot Apex? (In Plain English)

PocketBot Apex is an automated trading program that lives on your Windows computer. Instead of you sitting in front of blinking price charts guessing when to buy or sell, PocketBot Apex connects to global trading platforms (**MetaTrader 5** and optionally **cTrader**), watches **10 major markets at the same time**, tests every trade 10,000 times using computer math before risking a cent, and manages your Stop-Loss and Take-Profit safety lines automatically.

- **100% Safe Fake-Money Practice Default:** Out of the box, PocketBot Apex is physically locked to **Demo Mode** (virtual practice money). Even if you accidentally log into a real-money account, the bot will block itself and refuse to risk real cash unless two secret unlock switches are manually turned on.
- **$0 Monthly Cost to Run:** Running PocketBot Apex every day uses free Python and free MetaTrader 5 (plus optional free cTrader). You do NOT need to pay any monthly subscription to run the bot.
- **Zero Coding Needed:** Inside your folder, you get numbered **double-click buttons** (0 through 6) that install, test, launch, and configure everything for you.

2. What Your Computer Needs (System & Account Checklist)

Item	Minimum Needed	Plain-English Reason
Computer Type	Windows 10 or Windows 11 (64-bit)	MetaTrader 5's Python connector ONLY runs on 64-bit Windows PCs/laptops. Will NOT work on Mac, Chromebook, iPad, or phones.
Memory & Space	4 GB RAM (8 GB+ ideal) & 2 GB free on C: drive	Needed so MetaTrader 5 and the 10,000-simulation math engine run smoothly without lag.
Internet	Steady Wi-Fi or Ethernet cable	Streams live market prices every second and sends your orders to the broker server.
Power & Sleep	Charger plugged in + Windows Sleep set to NEVER	CRITICAL: If your laptop screen sleeps or you close the lid, Windows freezes all apps and the bot stops watching markets.
Free Software	MetaTrader 5 + Python 3.12 (Optional: Node.js, agy, cTrader)	0_FIRST_TIME_SETUP.bat installs Python 3.12 automatically. Exact manual commands are also in Section 5.

Item	Minimum Needed	Plain-English Reason
Free Accounts	1. Free MetaTrader 5 Demo Account 2. Optional: Google Account / cTrader ID / Telegram	MT5 Demo takes 30 seconds (no ID or credit card). Google/cTrader/Telegram are optional power-ups explained in Section 8.

DO THIS FIRST — Turn Off Windows Sleep Mode Right Now (Takes 20 Seconds):
1. Click the **Windows Start** icon (bottom-left corner of your screen) and click **Settings** (the gear icon).
2. Click **System**, then click **Power & sleep** (or **Power & battery** on Windows 11).
3. Find **"When plugged in, put my device to sleep after"** and change it to **Never**.
4. Always keep your laptop plugged into its wall charger while PocketBot Apex is running!

3. How PocketBot Apex Thinks & Makes Money (With Real Math)

Every 5 minutes, PocketBot Apex checks 10 markets: **EURUSD, GBPUSD, USDJPY, XAUUSD (Gold), SPX (S&P 500), NQ (Nasdaq 100), BTCUSD (Bitcoin), ETHUSD (Ethereum), TSLA (Tesla), and NVDA (Nvidia)**. Before a single trade is allowed through, it must survive **6 strict engineering layers**:

Layer 1: The Market Regime Brain (Trend vs. Sideways Chop Detector)

Normal trading bots lose money because they keep buying breakouts when the market is flat and choppy. PocketBot Apex calculates a mathematical score called the **Hurst Exponent (H)** and **Efficiency Ratio (ER)**:

- **TRENDING_EXPANSION (H > 0.50):** Price is moving cleanly in one direction — trading is **GREEN-LIT**.
- **MEAN_REVERTING_CHOP (H < 0.50):** Price is bouncing back and forth in a tight box — breakouts are **AUTO-BLOCKED**.
- **VOLATILITY_SHOCK:** Crazy news just hit and broker fees spiked — trading is **LOCKED DOWN** until calm returns.

Layer 2: Two-Speed Scanner & Level 2 Order-Book X-Ray

- **Low-Fee Markets (Checked every 5 minutes — M5):** EURUSD, GBPUSD, USDJPY, Gold, SPX, and NQ are scanned for clean 5-minute Opening Range Breakouts.
- **High-Fee Fast Markets (Checked every 15 minutes — M15 Strong Momentum):** Brokers charge higher entry fees ('spreads') on Bitcoin, Ethereum, Tesla, and Nvidia. So PocketBot zooms out to 15-minute charts and demands a candle **1.5x bigger than normal** closing in the **top/bottom 25%** of its bar with fees under 15% of the Stop-Loss.
- **Hydra Dual-Broker Mesh (MT5 + cTrader):** Works on MT5 alone right away. If you also connect free cTrader later, both run together — reading live institucional buy/sell walls (Level 2 Order Book) and automatically routing each trade to whichever broker has the cheaper fee!

Layer 3: Time Machine v2 (10,000 Realistic Practice Replays)

Before risking $1, PocketBot takes the last **60 days** of real market bars and simulates your exact trade **10,000 times** (using Stationary Block Bootstrap math), subtracting exact broker commissions and slippage on every path. If the average result (**net_avg_R**) is zero or negative, the trade is thrown in the trash immediately.

Layer 4: Exact Profit Math — Why MetaTrader 5 Is NOT Like Binary Betting

CRITICAL MATH EXPLANATION — Read This Carefully!
If you have ever used binary-options websites (like Pocket Option binary) or sports betting apps, you might think "risking $50 to win double ($100)" means handing over $50 upfront and getting your own $50 back + $50 profit.
MetaTrader 5 and cTrader DO NOT WORK LIKE THAT!

- **You pay $0 upfront bet:** Your entire $10,000.00 stays in your account balance.
- **You get paid by PRICE DISTANCE (Pips/Points):** PocketBot automatically places your **Take-Profit (TP)** line **TWICE AS FAR AWAY** as your **Stop-Loss (SL)** line (called a **2R** or 2-to-1 ratio), and sizes your trade ('Lots') so your worst-case Stop-Loss never exceeds **0.5% of your account ($50 on $10,000)**.
- **Real Example on $10,000 (Buying 0.25 Lots of EURUSD = $2.50 per pip step):**
  1. If price drops **20 pips** to Stop-Loss: 20 × -$2.50 = **-$50.00 pure loss** (Balance becomes **$9,950.00**).
  2. If price rises **40 pips** to Take-Profit (twice as far!): 40 × +$2.50 = **+$100.00 pure profit** added on top of your $10,000 (Balance becomes **$10,100.00**)!

Outcome on $10,000 Account	Price Distance	Pure Dollar Result	Out of 10 Trades	Total Impact
Losing Trade (Hits Stop-Loss)	20 pips against you	-$50.00 pure loss	6 losses (Wrong 60%!)	6 x -$50 = -$300.00
Winning Trade (Hits Take-Profit)	40 pips your way (2x)	+$100.00 pure profit	4 wins (Right only 40%!)	4 x +$100 = +$400.00
FINAL NET ACCOUNT BALANCE	—	—	**10 total trades**	**$10,000 - $300 + $400 = $10,100 (+$100 Profit!)**

Layer 5 & 6: Hybrid 3-Tier AI Veto, 6 Safety Brakes & Self-Healing Quarantine

- **Hybrid 3-Tier AI Router:** Automatically uses Google AI Studio API or Google AI Pro (agy) if set up on your PC to double-check for macro traps — and if you have NEITHER set up, it automatically runs in **100% Pure Math Mode** with zero errors!
- **6 Hard Circuit Breakers:** (1) Demo-Only Lock, (2) -2% Daily Loss Kill-Switch (-$200 max daily loss on $10k), (3) Max 2 Open Trades + Currency Correlation Shield (stops you from accidentally betting 3x against the US Dollar across EURUSD + GBPUSD + Gold at once!), (4) 15% Max Fee/Spread Cap, (5) Level 2 Order-Book Wall Guard, and (6) Broker-Stored Stop-Loss & Take-Profit (protects open trades even if your Wi-Fi or PC dies!).
- **Self-Healing Quarantine:** Saves every trade into a database (apex_ledger.db). If any symbol starts losing over 15–20 trades, PocketBot automatically quarantines that symbol into Paper Practice Mode until it recovers!

4. Complete Step-by-Step Setup From Zero (Every Click & Key)

Follow Steps 1 through 4 in exact order. Do NOT skip a step!

Step 1: Copy PocketBot_Apex From the USB Pendrive Onto Your C: Drive

Never run the bot directly off the USB stick (if the USB bumps loose, the bot crashes).

- 1. Plug the USB pendrive into your Windows computer.
- 2. Click the yellow **File Explorer** folder icon at the bottom of your Windows screen.
- 3. Click your **USB Drive** on the left side of the window.
- 4. Right-click the **PocketBot_Apex** folder and click **Copy**.
- 5. Click **This PC** on the left side, then double-click **Local Disk (C:)**.
- 6. Right-click an empty white space inside C: and click **Paste**. Wait for copying to finish.
- 7. Double-click your new **C:\PocketBot_Apex** folder to confirm you see the numbered buttons inside (starting with **0_FIRST_TIME_SETUP**).

Step 2: Install MetaTrader 5 & Open Your Free Practice Account

- 1. Open your internet browser (Edge or Chrome), type **https://www.metatrader5.com/en/download** in the top address bar, and press **Enter**.
- 2. Click the blue **Download MetaTrader 5 for Windows** button.
- 3. Open the downloaded file (**mt5setup.exe**), click **Yes** if Windows asks permission, and click **Next** then **Finish** until MetaTrader 5 opens.
- 4. **Open a Bot-Friendly Free Demo Account:**
  - At the very top-left corner of MetaTrader 5, click **File** → **Open an Account**.
  - Type **MetaQuotes** in the search box, click **Find your company**, click **MetaQuotes Ltd.**, and click **Next**.
  - Select **"Open a demo account to trade virtual money without risk"** and click **Next**.
  - Type your name, email, and phone number (no credit card or ID needed), check **"I agree..."**, and click **Next** → **Finish**.

  (WARNING: Do NOT use Pocket Option's PoTrade-MT5 server — Pocket Option blocks automated bots with Error 10026! MetaQuotes-Demo or Fusion Markets MT5 allow bots immediately.)
- 5. **Turn ON the Green 'Algo Trading' Switch (CRITICAL):**
  - Click **Tools** at the top menu of MetaTrader 5 → click **Options**.
  - Click the **Expert Advisors** tab at the top of that small popup box.
  - Check the box **"Allow algorithmic trading"** and click **OK**.
  - Look at the top toolbar of MetaTrader 5: confirm the **Algo Trading** button has a **GREEN play triangle** (if it is a red square, click it once so it turns green).
- 6. Press **Ctrl + T** on your keyboard so the **Trade** tab appears at the bottom showing your fake Demo Balance.
- 7. Click the minimize line (_) at the top-right of MetaTrader 5 — **DO NOT click X to close it!** MetaTrader 5 must stay open in the background whenever PocketBot runs.

Step 3: Double-Click '0_FIRST_TIME_SETUP' (Automatic 1-Click Installer)

- 1. Open File Explorer to **C:\PocketBot_Apex**.
- 2. Double-click **0_FIRST_TIME_SETUP** (or **0_FIRST_TIME_SETUP.bat**).

- 3. If Windows shows a blue box saying *"Windows protected your PC"*, click **More info** (underlined text) and click **Run anyway**.
- 4. If Windows asks *"Do you want to allow this app to make changes?"* while installing Python 3.12, click **Yes**.
- 5. Wait 2 to 4 minutes while the black window automatically installs Python 3.12, builds your private **.venv** engine, installs all math/trading libraries, tests your MetaTrader 5 connection, and prints **SETUP COMPLETE!**

Step 4: Manual PowerShell Backup Setup (ONLY If You Prefer Typing Commands)

What is Windows PowerShell? PowerShell is Windows' built-in blue/black text command box. You only need this if you ever want to run setup or commands by hand instead of using the double-click buttons:

- 1. Press the **Windows key** on your keyboard (bottom-left, next to Ctrl and Alt), type **PowerShell**, and click **Windows PowerShell**.
- 2. **How to Paste & Run:** Highlight any gray box below, press **Ctrl + C** to copy, click once inside PowerShell, press **Ctrl + V** (or right-click) to paste, and press **Enter**.
- 3. Unlock scripts and install Python 3.12 + Node.js + Antigravity AI CLI (paste one line at a time, press Enter, type **Y** + Enter if asked):

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
winget install -e --id Python.Python.3.12
winget install -e --id OpenJS.NodeJS.LTS
irm https://antigravity.google/cli/install.ps1 | iex
```

- 4. **IMPORTANT:** Click **X** to close PowerShell completely, open a **brand-new Windows PowerShell** window, and paste this block to build your Python engine:

```
cd C:\PocketBot_Apex
& "$env:LOCALAPPDATA\Programs\Python\Python312\python.exe" -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install MetaTrader5 numpy scipy ctrader-open-api twisted pyOpenSSL
service_identity aiohttp python-dotenv rich pytest
```

5. Daily Use: Your Numbered 1-Click Buttons Explained

Once setup is done, **you NEVER need PowerShell or coding open to use PocketBot Apex!** Just keep MetaTrader 5 minimized in the background (with green Algo Trading ON), open **C:\PocketBot_Apex**, and double-click:

Button Name in C:\PocketBot_Apex	What Happens When You Double-Click It
0_FIRST_TIME_SETUP.bat	Run ONCE on a new PC. Installs Python 3.12, builds .venv, installs all libraries, and checks MT5.
1_SETUP_CTRADER_KEYS.bat	Optional. Connects a cTrader account (once Spotware approves your app) to unlock Level 2 Order Book + Dual-Broker routing.
2_START_APEX_AUTO_DEMO.bat	**MAIN FULL-AUTO BUTTON:** Scans all 10 markets every 5 minutes, checks Regime + 10,000 Time Machine replays + 6 Safety Brakes, and places Demo trades automatically.
3_START_APEX_CONFIRM.bat	**MANUAL CO-PILOT BUTTON:** Runs the exact same analysis, shows you the trade ticket & odds on screen, and waits for you to type **YES** or **NO** + Enter before placing any trade.
4_APEX_LIVE_DASHBOARD.bat	**LIVE COLOR DASHBOARD:** Opens the live visual table showing your Balance, Daily Profit/Loss, every symbol's Trend/Chop Regime, 10,000-replay EV score, AI status, and Quarantine health.
5_CONFIGURE_AI_MODE.bat	**AI SWITCHER MENU:** Press 1–5 to pick AI priority (AI Studio API vs Google AI Pro vs Pure Math Only) or paste an API key.
6_CONFIGURE_BROKER_MESH.bat	**BROKER MESH MENU:** Press 1–5 to switch between Auto Hydra (MT5+cTrader), cTrader Eyes + MT5 Hands, 50/50 Split, or run a live broker fee shootout.

How to Stop the Bot or Close a Trade Manually:

- **Stop scanning for new trades:** Click the **X** at the top-right of the black PocketBot window (or hold **Ctrl** and tap **C**).
- **Your open trades stay safe:** Closing PocketBot does NOT erase your Stop-Loss or Take-Profit — both live on the broker's server.
- **Close a trade immediately by hand:** Open MetaTrader 5, look at the bottom **Trade** tab, and click the tiny **X** at the far right end of that trade's row.

6. How to Read the Live Apex Dashboard Screen

Dashboard Column	Plain-English Meaning	What Good Looks Like
Tier / TF	Chart speed used (TIGHT M5 = 5-min; WIDE M15 = 15-min)	Automatic per symbol
Best Venue	Cheapest broker right now + total fee in pips	MT5 or CTRADER (lowest pips wins)

Dashboard Column	Plain-English Meaning	What Good Looks Like
Market Regime	Live market state (Trending vs Sideways Chop vs Shock)	TRENDING_EXPANSION (Chop is blocked)
Hurst / ER	Trend math score (H > 0.50 = Trending; H < 0.50 = Chop)	H = 0.52+ and higher ER
L2 DOM OBI	Buyer vs Seller wall pressure (active when cTrader linked)	+0.15+ for Buy / -0.15- for Sell
TM net_avg_R	Average profit across 10,000 replays AFTER all fees	Positive (+0.10R to +1.00R+)
Q-Kelly Risk	Auto-calculated account risk % for this trade	0.10% to 0.50% max (0.00% = skipped)
AI Veto	AI Risk Committee decision (or IDLE / PURE_MATH)	APPROVE (VETO blocks risky traps)
Ledger State	Self-healing monitor (ACTIVE vs PAPER_SHADOW)	ACTIVE (PF = Profit Factor)

7. Complete PowerShell Commands & How to Use Antigravity (agy)

Do you need PowerShell or Antigravity (agy)? NO! The 1-click buttons above do everything. Use Section 7 ONLY if you want to run manual tests or talk to the built-in AI mechanic (**agy**) in plain English to customize settings.

Part A: Direct Manual PowerShell Commands

Open Windows PowerShell and ALWAYS run these two lines first to enter PocketBot Apex:

```
cd C:\PocketBot_Apex
.\.venv\Scripts\Activate.ps1
```

What You Want to Do	Exact Command to Paste Into PowerShell
Run all 25 self-diagnostic safety tests	python -m pytest tests/ -v
Open the live Rich color dashboard	python -m apex.cockpit.tui_dashboard
Run 1 single dry-run scan (no orders sent)	python apex_runner.py --mode demo-auto --once --dry-run
Run Full Auto Demo Mode	python apex_runner.py --mode demo-auto
Run Manual YES/NO Confirmation Mode	python apex_runner.py --mode confirm
Run Shadow A/B Filter Comparison Mode	python apex_runner.py --mode shadow-ab
Open the AI Mode Switcher Menu	python tools/configure_ai_mode.py
Open the Broker Mesh Switcher Menu	python tools/configure_broker_mesh.py

Part B: How to Open & Talk to Antigravity CLI (agy — Your AI Mechanic)

- 1. Open **Windows PowerShell**, paste **cd C:\PocketBot_Apex** + Enter, then **.\.venv\Scripts\Activate.ps1** + Enter.

- 2. Type **agy** and press **Enter**. *(If PowerShell says 'agy' is not recognized, paste **& "$env:LOCALAPPDATA\agy\bin\agy.exe"** and press Enter. If not installed yet, run **irm https://antigravity.google/cli/install.ps1 | iex** first.)*
- 3. Press **Enter** to trust the folder. Your web browser will pop open — sign in with your **Google / Gmail account** and click **Allow**.
- 4. Next to the blinking **>** at the bottom of PowerShell, type any plain-English request and press **Enter** (press **Enter** whenever agy asks permission):
  - *"Check my open MetaTrader 5 broker symbols, map any ticker differences, and run a dry-run test."*
  - *"Change max risk per trade from 0.5% to 0.25% and verify all tests pass."*
  - *"Read trader/apex_ledger.db and explain today's scans and trades in simple English."*
- 5. **To exit agy:** Hold **Ctrl** and tap **C** twice (**Ctrl + C, Ctrl + C**) or type **/quit** + Enter.

8. Optional Power-Ups (AI Modes, cTrader & Phone Panic Button)

Power-Up A: Switching AI Modes (AI Studio vs. Google AI Pro vs. Pure Math)

- **Zero-Setup Default:** If you have neither Google AI Pro nor an API key, PocketBot Apex automatically runs in **Pure Math Mode** — you don't have to do anything!
- **Free Google AI Studio Key (Optional):** Go to **https://aistudio.google.com/apikey**, sign in with Google, click **Create API key**, copy it, double-click **5_CONFIGURE_AI_MODE.bat**, press **4**, and paste the key.
- **Google AI Pro Plan (Optional):** If you pay for Google AI Pro and signed into **agy**, PocketBot Apex detects it automatically with zero keys!

Power-Up B: Connecting cTrader (Level 2 Order Book + Dual-Broker Routing)

- 1. Open a free Demo account with a cTrader broker (such as **Fusion Markets** — available in most countries and Canadian provinces *except Ontario & Alberta* — or directly inside **https://ctrader.com**).
- 2. Go to **https://openapi.ctrader.com/apps**, sign in with your cTrader ID, click **Add new app**, add Redirect URI **http://127.0.0.1:8787/callback**, and save.
- 3. **IMPORTANT WAIT NOTE:** New cTrader apps start with Status: **Submitted** (orange clock) while Spotware reviews KYC/activation (takes a few hours to 1 business day). Keep running on MT5 until Status turns green **Active** — then double-click **1_SETUP_CTRADER_KEYS.bat**, paste your Client ID & Secret, and Hydra Active-Active Mode unlocks automatically!

Power-Up C: Phone Alerts & 1-Tap '/panic_flatten' Emergency Kill-Switch

- 1. Install the free **Telegram** app on your phone, search for **@BotFather** (blue checkmark), tap **Start**, send **/newbot**, follow the 2 prompts, and copy the **HTTP API Token**.
- 2. Send **hello** to your new bot, then search Telegram for **@userinfobot**, tap **Start**, and copy your numeric **Id**.
- 3. Open **C:\PocketBot_Apex\.env** in Notepad (or create it), add **TELEGRAM_BOT_TOKEN=your_token** and **TELEGRAM_CHAT_ID=your_id**, and save. You can now send **/status**, **/pause**, or **/panic_flatten** (closes all open trades immediately!) from your phone anywhere in the world.

9. Troubleshooting Common Errors & Honest Rules to Make Money

Problem / Message You See	Exact 20-Second Fix
Error 10026: AutoTrading disabled by server	Your MT5 broker (e.g., Pocket Option) blocks bots on demo. In MT5 click **File** → **Open an Account** → **MetaQuotes Ltd.** (or Fusion Markets MT5) to open a bot-friendly Demo account. (Meanwhile, PocketBot safely tracks Paper-Shadow trades automatically.)
Error 10027: AutoTrading disabled by client	Look at the top toolbar of MetaTrader 5 — the **Algo Trading** button is red. Click it once so it turns **GREEN**.
MT5 Connection FAILED / Offline	MetaTrader 5 is closed or logged out. Open MetaTrader 5, confirm your Demo Balance shows at the bottom, and minimize it (never close it).
cTrader says 'Submitted' / KYC pending	Normal for brand-new Spotware Open API apps. Leave cTrader unset for today — PocketBot Apex runs 100% on MT5 automatically until Spotware flips Status to Active.

Problem / Message You See	Exact 20-Second Fix
Windows says 'Windows protected your PC'	Normal for local .bat launcher files. Click **More info** then click **Run anyway**.
Black window flashes and closes instantly	You haven't run **0_FIRST_TIME_SETUP.bat** yet on this PC (or you ran it directly off the USB stick). Copy the folder to **C:\PocketBot_Apex** and run **0_FIRST_TIME_SETUP.bat** first.

The 3 Golden Rules Before Ever Touching Real Money:

- **1. Best Hours to Run It (Eastern / New York Time — ET):** Run during **8:00 AM – 12:00 PM ET Monday–Friday** (London + New York overlap) when EURUSD, Gold, SPX, and NQ move cleanest. Global Forex, Gold, and Stock markets close Friday 5:00 PM ET to Sunday 5:00 PM ET (only Crypto moves on weekends if supported by your broker).
- **2. Honest Limits — Losing Streaks Happen:** No bot wins every trade. Losing 3 to 5 trades in a row is normal math. Never judge the bot on 1 or 2 days, and remember fast surprise news can jump slightly past a Stop-Loss ('slippage').
- **3. The 50-Trade Graduation Rule:** Keep running strictly on **Demo Fake Money** until you have logged at least **50 closed trades** with a **Profit Factor above 1.30** ($1.30+ won per $1.00 lost). If you ever choose to trade live later, use a regulated broker in your country, start tiny ($200–$500 at 0.01 lots), and NEVER risk rent, bill, or essential money.

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